

Case Study #3: Forecasting with AR and ARIMA Models

Case Solutions

Consider the quarterly data on Walmart revenues (in \$million) from the first quarter (Q1) of 2006 through the third quarter (Q3) of 2025 (the same data set as in case study #2, *673_case2.csv*). The goal is to forecast Walmart's quarterly revenue in the fourth quarter (Q4) of 2025 and Q1-Q4 of 2026-2027.

As you did in *case study #2*, start this case with the following: Create time series data set in R using the *ts()* function (this part will not be graded in case #3).

Develop data partition with the validation partition of 19 periods and the rest for the training partition (this part will not be graded in case #3).

```
> train.ts
      Qtr1   Qtr2   Qtr3   Qtr4
2006 79676  85430  84467  98795
2007 86410  92999  91865 105749
2008 94940 102342  98345 108627
2009 94242 100876  99373 113594
2010 99811 103726 101952 116360
2011 104189 109366 110226 122728
2012 113010 114282 113800 127559
2013 114070 116830 115688 129706
2014 114960 120125 119001 131565
2015 114826 120229 117408 129667
2016 115904 120854 118179 130936
2017 117542 123355 123179 136267
2018 122690 128028 124894 138793
2019 123925 130377 127991 141671
2020 134622 137742 134708 152079
```

```
> valid.ts
      Qtr1   Qtr2   Qtr3   Qtr4
2021 138310 141048 140525 152871
2022 141569 152859 152813 164048
2023 152301 161632 160804 173388
2024 161508 169335 169588 180554
2025 165609 177402 179496
```

1. Identify time series predictability.

1a. Using the *AR(1)* model for the historical data, Provide and explain the *AR(1)* model summary in your report. Explain if the Walmart revenue is predictable.

The output of the *AR(1)* model for *revenue.ts* time series data is presented below. *ARIMA(1, 0, 0)* is an autoregressive (AR) model with order 1, no differencing, and no moving average model.

```
Series: revenue.ts
ARIMA(1,0,0) with non-zero mean
```

```
Coefficients:
      ar1      mean
  0.9485 127211.70
s.e.  0.0399  17576.13
```

```
sigma^2 = 96984434: log likelihood = -838.64
AIC=1683.28  AICc=1683.6  BIC=1690.39
```

```
Training set error measures:
      ME      RMSE      MAE      MPE      MAPE      MASE      ACF1
Training set 966.264 9722.609 8265.906 0.2226992 6.735426 1.69258 -0.6625217
```

The model's equation is:

$$Y_t = 127211.7 + 0.9485 Y_{t-1}$$

The coefficient of the *ar1* (Y_{t-1}) variable, $\beta_1 = 0.9485$, and standard error of estimate, $s.e. = 0.0399$. We will use these two parameters for hypothesis testing about the value of the AR(1) regression coefficient.

Hypothesis Testing: Z- Test

Null hypothesis $H_0: \beta_1 = 1$

Alternative hypothesis $H_1: \beta_1 \neq 1$

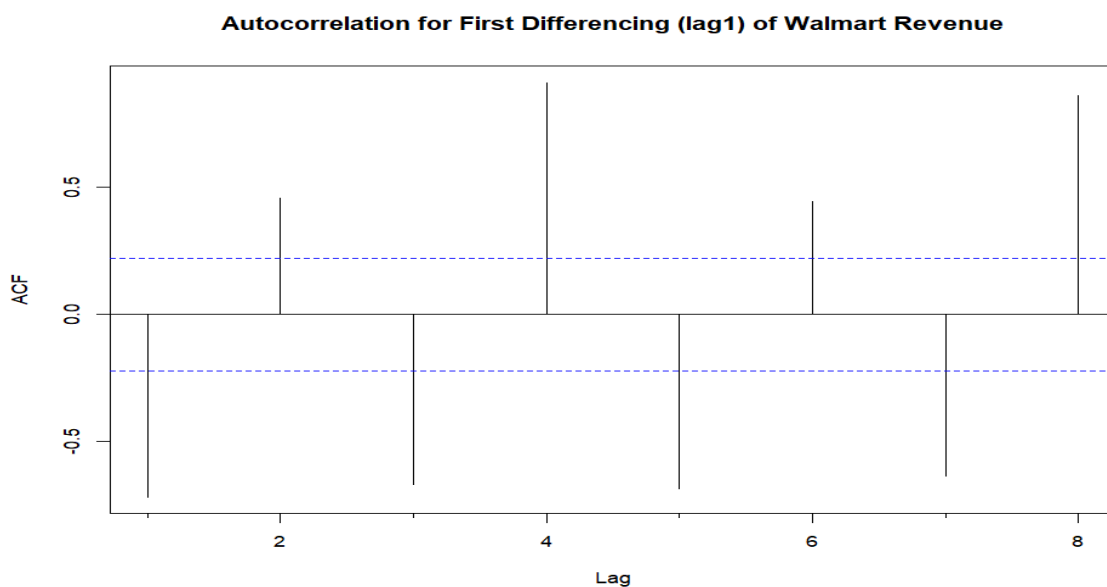
z-statistic = $(\beta_1 - 1)/(s.e.) = (0.9485 - 1)/0.0399 = -1.1528$

p-value for z-statistic = 0.0984

Based on the p-value of 0.0984, which is greater than 0.05, we cannot reject (need to accept) the null hypothesis that $\beta_1 = 1$. Therefore, the time series data for Walmart revenue, *revenue.ts*, according to this test, is not predictable and is considered random walk.

1b. Using the first differencing (lag-1) of the historical data and *Acf()* function, provide in the report the autocorrelation plot of the first differencing (lag-1) with the maximum of 8 lags and explain if Walmart revenue is predictable.

The autocorrelation plot of the first differencing for the *revenue.ts* data is presented below.



All autocorrelation coefficients of the first (once) differenced data are statistically significant, in particular, in lag-1 for trend and lag-4 for quarterly seasonality. Therefore, using the first differencing, we can confirm that *revenue.ts* is not random walk and is predictable. Because the results of the two predictability tests in 1b and 1c are opposite, we will continue to utilize the data set as predictable in forecasting Walmart revenue in Q4 of 2025 and Q1-Q4 of 2026-2027.

2. Apply the two-level forecast with regression model and AR model for residuals.

2a. For the training data set, use the `tslm()` function to develop a *regression model with linear trend and seasonality*. Forecast Walmart's revenue with the `forecast()` function (use the associated R code from case #2). *No explanation is required in your report.*

The output for the regression model with linear trend and seasonality for the training period and for ecast for the validation period are shown below (not graded in this case; were used in case #2).

```
Call:
tslm(formula = train.ts ~ trend + I(trend^2) + season)

Residuals:
    Min       1Q   Median       3Q      Max
-4914.3 -1858.3  -600.4  1937.2  7673.1

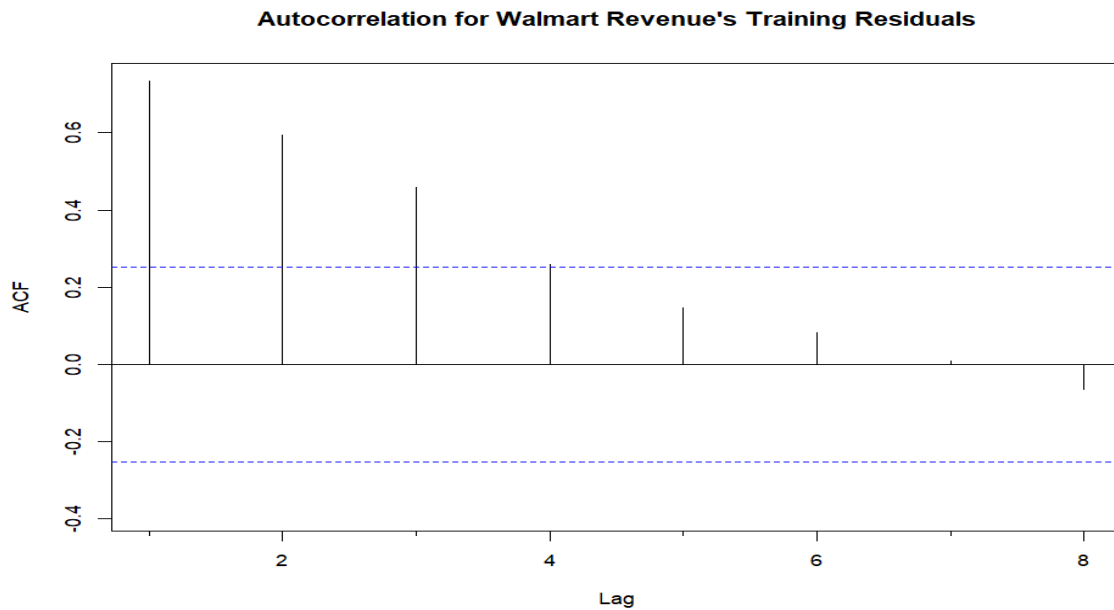
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  81899.443   1338.764   61.175 < 2e-16 ***
trend        1161.865     89.516   12.979 < 2e-16 ***
I(trend^2)    -6.030       1.422   -4.240 8.8e-05 ***
season2       4243.516   1078.843    3.933 0.000241 ***
season3       1750.491   1079.520    1.622 0.110726
season4      14503.194   1080.643   13.421 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2954 on 54 degrees of freedom
Multiple R-squared:  0.9672, Adjusted R-squared:  0.9641
F-statistic: 318.3 on 5 and 54 DF, p-value: < 2.2e-16
```

```
> train.trend.season.pred
      Point Forecast      Lo 0      Hi 0
2021 Q1    130334.9 130334.9 130334.9
2021 Q2    134998.6 134998.6 134998.6
2021 Q3    132913.7 132913.7 132913.7
2021 Q4    146062.4 146062.4 146062.4
2022 Q1    131943.2 131943.2 131943.2
2022 Q2    136558.6 136558.6 136558.6
2022 Q3    134425.4 134425.4 134425.4
2022 Q4    147525.9 147525.9 147525.9
2023 Q1    133358.4 133358.4 133358.4
2023 Q2    137925.6 137925.6 137925.6
2023 Q3    135744.2 135744.2 135744.2
2023 Q4    148796.5 148796.5 148796.5
2024 Q1    134580.8 134580.8 134580.8
2024 Q2    139099.7 139099.7 139099.7
2024 Q3    136870.1 136870.1 136870.1
2024 Q4    149874.1 149874.1 149874.1
2025 Q1    135610.1 135610.1 135610.1
2025 Q2    140080.8 140080.8 140080.8
2025 Q3    137802.9 137802.9 137802.9
```

2b. Identify the regression model's residuals for the training period and use the `Acf()` function with a maximum of 8 lags to identify autocorrelation for these residuals. Provide the autocorrelation plot in your report and explain if it would be a good idea to add to your forecast an AR model for residuals.

The autocorrelation chart (correlogram) of the residuals from the regression model with linear trend and seasonality (question 2a) is provided below.



The chart shows significant autocorrelation of residuals in lags 1-7, which means that these autocorrelations (relationships) between residuals are not incorporated into the regression model. Thus, modeling these residual autocorrelations with an *AR* model and developing a two-level model may improve the forecast.

2c. Develop an *AR(1)* model for the regression residuals, present and explain the model and its equation in your report. Use the *Acf()* function for the residuals of the *AR(1)* model (residuals of residuals), present the autocorrelation chart, and explain it in your report.

The output of the *AR(1)* model for regression residuals is presented below. *ARIMA(1, 0, 0)* is an autoregressive (AR) model with order 1, no differencing, and no moving average model.

Series: train.trend.season\$residuals
ARIMA(1,0,0) with non-zero mean

Coefficients:

	ar1	mean
	0.8429	321.0298
s.e.	0.0789	1292.2419

sigma^2 = 2970217: log likelihood = -531.86
AIC=1069.73 AICc=1070.16 BIC=1076.01

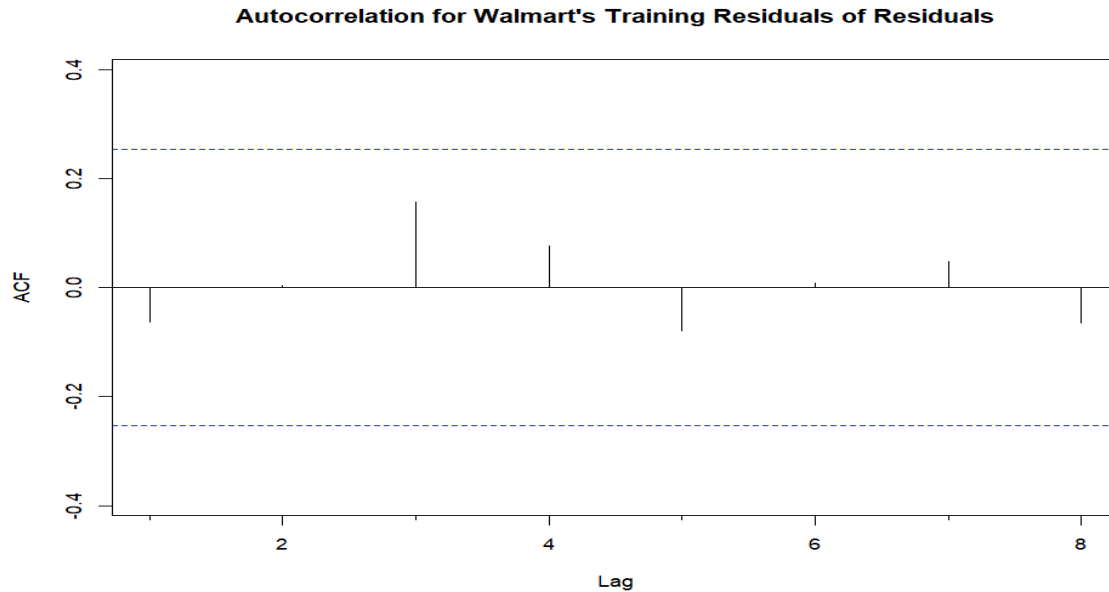
Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	81.31996	1694.465	1151.531	42.21883	82.14699	0.4754662	-0.06318279

The *AR(1)* model's equation is:

$$e_t = 321.0298 + 0.8429 e_{t-1}$$

An autocorrelation chart for the $AR(1)$ model's residuals (residuals of residuals) is presented below.



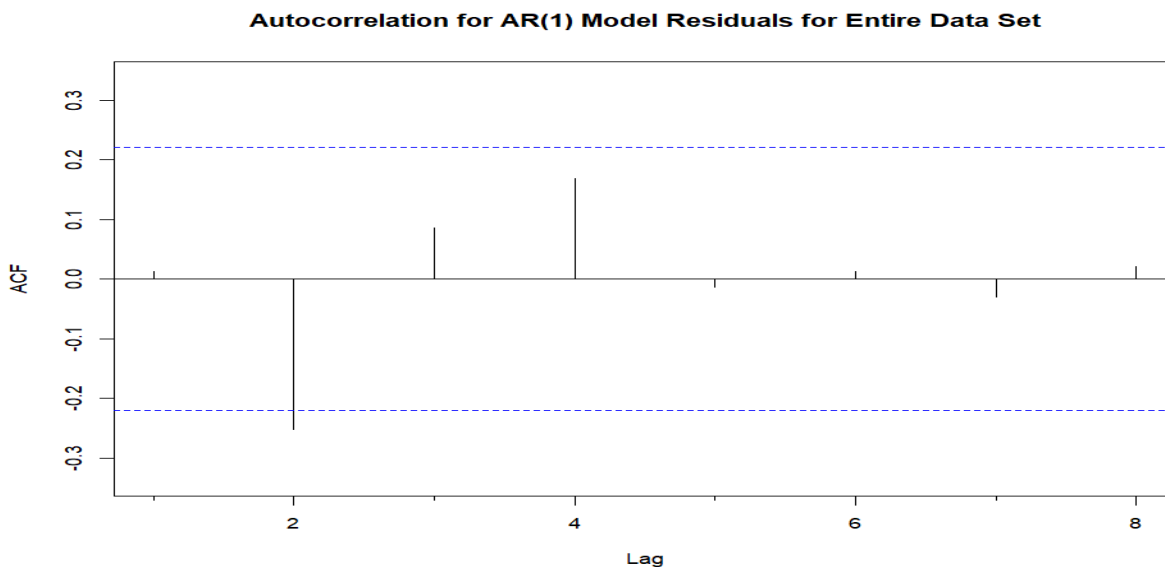
As can be seen from the autocorrelation chart (correlogram), all autocorrelations of residuals of residuals created by $AR(1)$ model are random (statistically insignificant). Thus, the $AR(1)$ model for residuals has absorbed significant autocorrelation in all lags. Therefore, the $AR(1)$ model for residuals can be combined with the regression model in question 2a to improve the time series forecast with the two-level forecasting model.

2d. Create a two-level forecasting model (regression model with *quadratic trend and seasonality* + $AR(1)$ model for residuals) for the validation period. Show in your report a table with the validation data, regression forecast for the validation data, $AR(1)$ forecast for the validation data, and combined forecast for the validation period.

The table below describes the revenue data and forecasts in the validation partition of 16 quarters in 2020-2023 (*Valid.Revenue*), regression model's forecast in the validation period (*Reg.Forecast*), $AR(1)$ model's forecast of the regression residuals in the validation period (*$AR(1)$ Forecast*), and combined forecast (*Combined.Forecast*) as a sum of the regression and $AR(1)$ models' forecasts.

	Valid.Revenue	Reg.Forecast	$AR(1)$ Forecast	Combined.Forecast
1	138310	130334.9	6517.8402	136852.8
2	141048	134998.6	5544.1066	140542.7
3	140525	132913.7	4723.3803	137637.0
4	152871	146062.4	4031.6185	150094.0
5	141569	131943.2	3448.5565	135391.7
6	152859	136558.6	2957.1137	139515.7
7	152813	134425.4	2542.8936	136968.3
8	164048	147525.9	2193.7618	149719.7
9	152301	133358.4	1899.4908	135257.9
10	161632	137925.6	1651.4599	139577.1
11	160804	135744.2	1442.4032	137186.6
12	173388	148796.5	1266.1966	150062.7
13	161508	134580.8	1117.6781	135698.4
14	169335	139099.7	992.4971	140092.2
15	169588	136870.1	886.9863	137757.0
16	180554	149874.1	798.0549	150672.1
17	165609	135610.1	723.0977	136333.2
18	177402	140080.8	659.9189	140740.7
19	179496	137802.9	606.6677	138409.6

2e. Develop a two-level forecast (regression model with *quadratic trend and seasonality* and *AR(1)* model for residuals) for the *entire data set*. Present in your report the autocorrelation chart for the *AR(1)* model's residuals and explain it. Also, provide a data table with the models' forecasts for Walmart revenue in Q4 of 2025 and Q1-Q4 of 2026-2027 (regression model, *AR(1)* for residuals, and two-level combined forecast).



The autocorrelation chart above of the *AR(1)* model residuals (residuals of residuals) shows that almost all autocorrelations are random (within horizontal thresholds) with exception of one autocorrelation coefficient for lag-2, which is also very close to the threshold (a weak autocorrelation). This means that the *AR(1)* model absorbed significant autocorrelations in the residuals.

The table below provides 9 forecasts for Q4 of 2025 and Q1-Q4 of 2026-2027, that are associated with: the regression model with linear trend and seasonality (*Reg.Forecast*), *AR(1)* model for the regression residuals (*AR(1)Forecast*), and two-level combined forecast (*Combined.Forecast*) as a sum of the regression and *AR(1)* models' forecasts.

	Reg.Forecast	AR(1)Forecast	Combined.Forecast
1	183176.1	9002.310	192178.5
2	170433.4	8453.146	178886.5
3	177006.3	7937.828	184944.2
4	176416.4	7454.271	183870.6
5	189744.7	7000.516	196745.2
6	177062.6	6574.727	183637.3
7	183696.2	6175.180	189871.3
8	183166.8	5800.258	188967.1
9	196555.8	5448.444	202004.3

3. Use ARIMA Model and Compare Various Methods.

3a. Use *Arima()* function to fit *ARIMA(1,1,1)(1,1,1)* model for the *training data set*. Present in your report the summary of this ARIMA model, show and briefly explain the *ARIMA* model and its equation in your report. Using this model, forecast revenue for the *validation period* and present it in your report.

The output from the *ARIMA(1,1,1)(1,1,1)* model for the training partition is presented below.

```
Series: train.ts
ARIMA(1,1,1)(1,1,1)[4]

Coefficients:
      ar1      ma1      sar1      sma1
    0.1418  -0.2677  0.2199  -1.0000
s.e.  0.7080   0.6814  0.1729   0.1623
```

```
sigma^2 = 3278090: log likelihood = -493.11
AIC=996.22 AICC=997.44 BIC=1006.26
```

Training set error measures:

```
ME RMSE MAE MPE MAPE MASE ACF1
Training set -179.1361 1669.244 1126.504 -0.186929 0.9681938 0.2837019 -0.0151062
```

This is a seasonal ARIMA model, $ARIMA(p, d, q)(P, D, Q)_m$, where:

- $p = 1$, order 1 autoregressive model $AR(1)$
- $d = 1$, first differencing
- $q = 1$, order 1 moving average $MA(1)$ for error lags
- $P = 1$, order 1 autoregressive model $AR(1)$ for the seasonal part
- $D = 1$, first differencing for the seasonal part
- $Q = 1$, order 1 moving average $MA(1)$ for the seasonal error lags
- $m = 4$, for quarterly seasonality.

The model's equation is:

$$y_t - y_{t-1} = 0.1418(y_{t-1} - y_{t-2}) - 0.2677e_{t-1} + 0.2199(y_{t-1} - y_{t-5}) - \rho_{t-1}$$

Using the model's equation, see below the forecast for the validation period:

Point	Forecast	Lo 0	Hi 0
2021 Q1	140086.5	140086.5	140086.5
2021 Q2	144630.0	144630.0	144630.0
2021 Q3	142619.7	142619.7	142619.7
2021 Q4	157061.6	157061.6	157061.6
2022 Q1	144407.9	144407.9	144407.9
2022 Q2	149325.0	149325.0	149325.0
2022 Q3	147548.4	147548.4	147548.4
2022 Q4	161347.3	161347.3	161347.3
2023 Q1	148548.3	148548.3	148548.3
2023 Q2	153547.6	153547.6	153547.6
2023 Q3	151822.4	151822.4	151822.4
2023 Q4	165479.8	165479.8	165479.8
2024 Q1	152648.9	152648.9	152648.9
2024 Q2	157666.2	157666.2	157666.2
2024 Q3	155952.4	155952.4	155952.4
2024 Q4	169578.7	169578.7	169578.7
2025 Q1	156740.8	156740.8	156740.8
2025 Q2	161762.1	161762.1	161762.1
2025 Q3	160050.7	160050.7	160050.7

3b. Use the `auto.arima()` function to develop an ARIMA model using the *training data set*. Present in your report the summary of this ARIMA model, show and explain the ARIMA model and its equation in your report. Use this model to forecast revenue in the *validation period* and present this forecast in your report.

The output from using the `auto.arima()` function for the training partition is presented below.

```
Series: train.ts
ARIMA(1,0,0)(1,1,0)[4] with drift
```

Coefficients:

```
ar1 sar1 drift
0.8233 -0.3287 1059.0613
s.e. 0.0854 0.1539 273.2764
```

```
sigma^2 = 4066426: log likelihood = -504.68
AIC=1017.36 AICC=1018.15 BIC=1025.46
```

Training set error measures:

```
ME RMSE MAE MPE MAPE MASE ACF1
Training set -51.98903 1895.26 1390.402 -0.05242751 1.211152 0.3501625 -0.02222372
```

This is a seasonal ARIMA model, $(1, 0, 0)(1, 1, 0)_4$, with the following parameters:

- $p = 1$, order 1 autoregressive model $AR(1)$
- $d = 0$, no differencing
- $q = 0$, no moving average model for error lags
- $P = 1$, order 1 autoregressive model for the seasonal part

- $D = 1$, first differencing for the seasonal part
- $Q = 0$, no moving average model for the seasonal error lags
- $m = 4$, for the quarterly seasonality.

The ARIMA model's equation (with drift) is:

$$y_t = 1059.0613 + 0.8233y_{t-1} - 0.3287(y_{t-1} - y_{t-5})$$

This ARIMA model's forecast in the validation period is the following:

	Point	Forecast	Lo 0	Hi 0
2021 Q1		141448.6	141448.6	141448.6
2021 Q2		144831.1	144831.1	144831.1
2021 Q3		141324.4	141324.4	141324.4
2021 Q4		156917.8	156917.8	156917.8
2022 Q1		146999.8	146999.8	146999.8
2022 Q2		149913.3	149913.3	149913.3
2022 Q3		146246.9	146246.9	146246.9
2022 Q4		162165.1	162165.1	162165.1
2023 Q1		151799.3	151799.3	151799.3
2023 Q2		154691.1	154691.1	154691.1
2023 Q3		150932.4	150932.4	150932.4
2023 Q4		166624.6	166624.6	166624.6
2024 Q1		156307.9	156307.9	156307.9
2024 Q2		159126.0	159126.0	159126.0
2024 Q3		155331.1	155331.1	155331.1
2024 Q4		171042.8	171042.8	171042.8
2025 Q1		160664.9	160664.9	160664.9
2025 Q2		163470.1	163470.1	163470.1
2025 Q3		159656.5	159656.5	159656.5

3c. Apply the *accuracy()* function to compare performance measures of the two ARIMA models in 3a and 3b. Present the accuracy measures in your report, compare them and identify, using MAPE and RMSE, the best ARIMA model to apply.

ARIMA Model (1,1,1)(1,1,1)₄

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	6043.977	8998.679	7568.489	3.498	4.557	0.758	0.929

Auto ARIMA (0,1,0)(0,1,1)₄

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	5061.382	8444.12	6871.769	2.887	4.15	0.61	0.871

Based on the MAPE accuracy measures, the best model is the auto ARIMA (1,0,0)(1,1,0)₄ model, which has the lowest values of MAPE (4.15%) and RMSE (8444.12) vs. the respective measures for the ARIMA model ARIMA(1,1,1)(1,1,1)₄. However, to make comparison for the entire data set, we would consider the latter ARIMA model as well.

3d. Use the two ARIMA models from 3a and 3b for the entire data set. Present models' summaries in your report. Use these ARIMA models to forecast Walmart revenue in Q4 of 2025 and Q1-Q4 of 2026-2027, and present these forecasts in your report.

ARIMA Model (1,1,1)(1,1,1)₄

The output for this ARIMA model for the entire data set is shown below.

```
Series: revenue.ts
ARIMA(1,1,1)(1,1,1)[4]

Coefficients:
      ar1      ma1      sar1      sma1
    0.4557  -0.5834  0.1373  -0.7321
s.e.  0.3834   0.3540  0.2637   0.2659

sigma^2 = 4623124: log likelihood = -671.98
AIC=1353.95  AICC=1354.84  BIC=1365.47
```

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	84.50456	2023.966	1424.491	0.0003819899	1.113934	0.2916879	0.04001782

$$y_t - y_{t-1} = 0.4557(y_{t-1} - y_{t-2}) - 0.5834e_{t-1} + 0.1373(y_{t-1} - y_{t-5}) - 0.7321\rho_{t-1}$$

The model's forecast for the 9 future quarters is the following:

	Point Forecast	Lo 0	Hi 0
2025 Q4	191149.4	191149.4	191149.4
2026 Q1	177684.9	177684.9	177684.9
2026 Q2	186649.9	186649.9	186649.9
2026 Q3	186967.1	186967.1	186967.1
2026 Q4	199078.3	199078.3	199078.3
2027 Q1	185982.7	185982.7	185982.7
2027 Q2	194634.7	194634.7	194634.7
2027 Q3	194742.3	194742.3	194742.3
2027 Q4	206932.0	206932.0	206932.0

Auto ARIMA Model

The output for this auto ARIMA model for the entire data set is shown below.

Series: revenue.ts
 ARIMA(0,1,0)(0,1,1)[4]

Coefficients:

smal	-0.6678
s.e.	0.1300

sigma^2 = 4552209: log likelihood = -672.93
 AIC=1349.85 AICC=1350.02 BIC=1354.46

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
Training set	80.31847	2050.968	1480.204	0.008590517	1.15597	0.3030961	-0.06502539

This auto ARIMA model's equation is:

$$y_t - y_{t-1} = -0.6678\rho_{t-1}$$

The model's forecast for the 8 future quarters is the following:

	Point Forecast	Lo 0	Hi 0
2025 Q4	191743.7	191743.7	191743.7
2026 Q1	178809.6	178809.6	178809.6
2026 Q2	187777.0	187777.0	187777.0
2026 Q3	188047.2	188047.2	188047.2
2026 Q4	200294.9	200294.9	200294.9
2027 Q1	187360.9	187360.9	187360.9
2027 Q2	196328.3	196328.3	196328.3
2027 Q3	196598.4	196598.4	196598.4
2027 Q4	208846.2	208846.2	208846.2

3e. Apply the *accuracy()* function to compare the performance measures of the following forecasting models for the *entire data set*: (1) regression model with *quadratic trend and seasonality*; (2) *two-level model* (with *AR(1)* model for residuals); (3) *ARIMA(1,1,1)(1,1,1)* model; (4) *auto ARIMA* model; and (5) *seasonal naïve* forecast for the entire data set. Present the accuracy measures in your report, compare them, and identify, using *MAPE* and *RMSE*, the best model to use for forecasting Walmart's revenue in Q4 of 2025 and Q1-Q4 of 2026-2027.

The accuracy measures for the 5 specified models (for the entire data set) are presented below.

Regression model with linear trend and seasonality

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	0	4733.615	4033.899	-0.15	3.328	0.863	0.484

Two-level model (with AR(1) model for residuals)

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	183.389	1974.071	1502.192	0.113	1.203	0.013	0.188

ARIMA model (1,1,1)(1,1,1)₄

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	84.505	2023.966	1424.491	0	1.114	0.04	0.194

Auto ARIMA model (0,1,0)(0,1,1)₄

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	80.318	2050.968	1480.204	0.009	1.156	-0.065	0.197

Seasonal naïve forecast

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test set	4729.24	5882.997	4883.613	3.652	3.789	0.735	0.555

According to the accuracy measures, the lowest MAPE of 1.11% and lowest RMSE of 2023.97 are for the *ARIMA (1,1,1)(1,1,1)₄* model. Based on the superiority of MAPE and RMSE, we should select the *ARIMA (1,1,1)(1,1,1)₄* model as the best model for forecasting in Q4 of 2025 and Q1-Q4 of 2026-2027.